Wavvar MMWK Module Documentation

ML6432A Module Introduction

Wavvar Technologies

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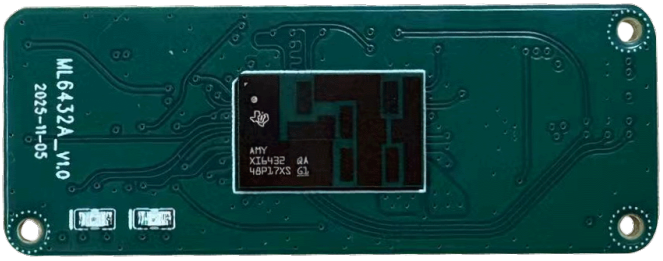
1. Module Introduction

ML6432A is a high-performance, low-power millimeter-wave radar module developed on the TI IWRL6432AOP chip. The module integrates the radar RF front end, the digital processing unit, and the antenna in a compact and highly integrated design. It is intended for applications such as smart homes, human presence detection, vital-sign monitoring, and motion detection. The module supports UART and SPI interfaces, making it convenient for rapid development and system integration.



ML6432A

2. Technical Specifications and Key Features

Category	Item	Specification	Remarks
Basic Parameters	Dimensions	15 × 39 × 7.2 mm	Module size, subject to the mechanical drawing
	Power Input	3.3V	Single power input, recommended power capability ≥1A
	Power Consumption	Depends on operating mode	Typical about 1W in Active mode; lower in low-power mode
	Communication Interfaces	UART / SPI / CAN-FD / SOP	Supports multiple interfaces for communication and boot configuration
Environmental Parameters	Operating Temperature	0 to +45°C	
	Storage Temperature	-55 to +150°C	
	Operating Humidity	≤95% (non-condensing)	
External Interfaces	P1	6-pin, SPI + UART communication interface	
	P2	6-pin, power + CAN + control interface	
P1 Interface Definition	SPIA_MOSI	SPI master transmit	
	SPIA_MISO	SPI master receive	
	SPIA_CS	SPI chip select	

Category	Item	Specification	Remarks
	SPIA_CLK	SPI clock	
	RS232_TX	UART transmit	TTL level
	RS232_RX	UART receive	TTL level
P2 Interface Definition	3.3V	Power input	
	GND	Power ground	
	NRESET	Module reset	Active low
	MCU_IO_MODE_SET	Mode control	Functional expansion interface
	CAN_FD_TX	CAN transmit	
	CAN_FD_RX	CAN receive	
Communication Capability	SPI	Configurable communication interface	
	UART	Main debugging / data communication	
	CAN	Supports CAN-FD	Industrial applications
RF Parameters	Operating Frequency Band	57 to 63.5 GHz	
	Tx/Rx Channels	2Tx 3Rx	
	Transmit Power	17 dBm	Typical EIRP per transmit channel (AOP, 0 dB back-off)
Detection Performance	Detection Range	0.1 to 20 m	Typical for human moving targets
	Micro-motion Detection	≤6 m	Respiration / stationary detection
	FOV	Typical horizontal ±70° / vertical ±60°	Depends on antenna and installation structure
Storage Function	Flash	On-board 16Mbit QSPI Flash	Approximately 2MB
Clock System	External Crystal	40 MHz	
Control and Status	Status Indicator	On-board LED	Network / running status
	Control Signals	NRESET / SOP / mode control	
On-board Features	Boot Mode	Flash / Application / Debug	Multiple boot modes
	Interface Type	Multi-interface multiplexing	SPI / UART / CAN

Category	Item	Specification	Remarks
	LED (Optional)	2x LED	

3. Application Areas

1 Healthcare monitoring: non-contact vital-sign monitoring (respiration and heart rate)

1 Building automation: automatic doors, occupancy detection, people tracking, and people counting

1 Consumer electronics: laptops, smart appliances (air conditioners, refrigerators, smart toilets), and smartwatches

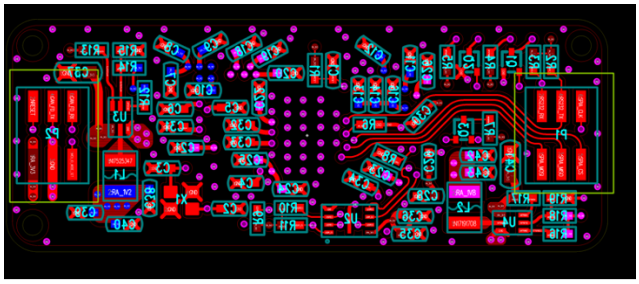
1 Security and surveillance: video doorbells, IP cameras, and motion detectors

1 Automotive electronics: in-cabin intrusion detection and related applications

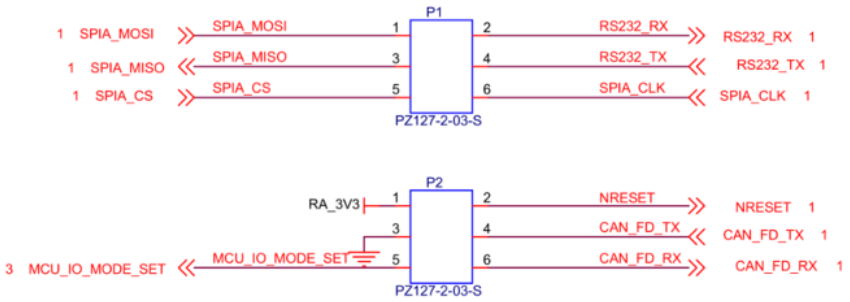
4. WDR/MDR Series Integration

In system-integration scenarios, `ML6432A` and `ML6432A_B0` share the same radar-side electrical interface definition, but they differ in the installation method. `ML6432A` is also supported by `MDR-M` / `WDR-M` at the functional level, but it usually connects through an adapter cable rather than by direct insertion into the controller board.

The images below come from the `WDR/MDR` integration material and can be used as supplementary references.



Connector layout and P1/P2 position references



Signal schematic reference from the WDR/MDR integration material

For the complete `WDR/MDR` system-level description, refer to mdr.md.

5. Interface Description

The module connects to external systems through two `6-pin` connectors (`P1` and `P2`). These interfaces include power, reset, mode configuration, and `SPI`, `UART`, and `CAN FD` communication signals. The current `ML6432A` interface references are shown below.

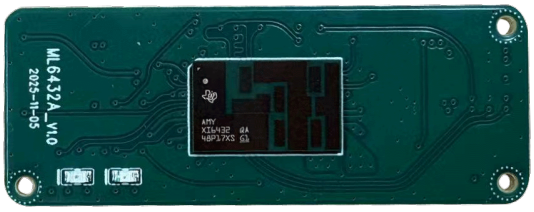


Figure 1. ML6432A front view

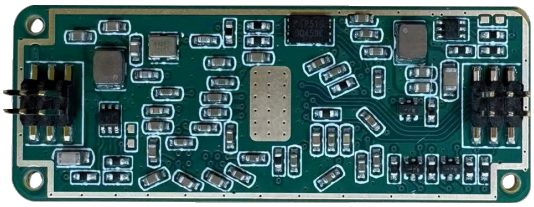


Figure 2. ML6432A back view

Figure 3. ML6432A connector pin assignment

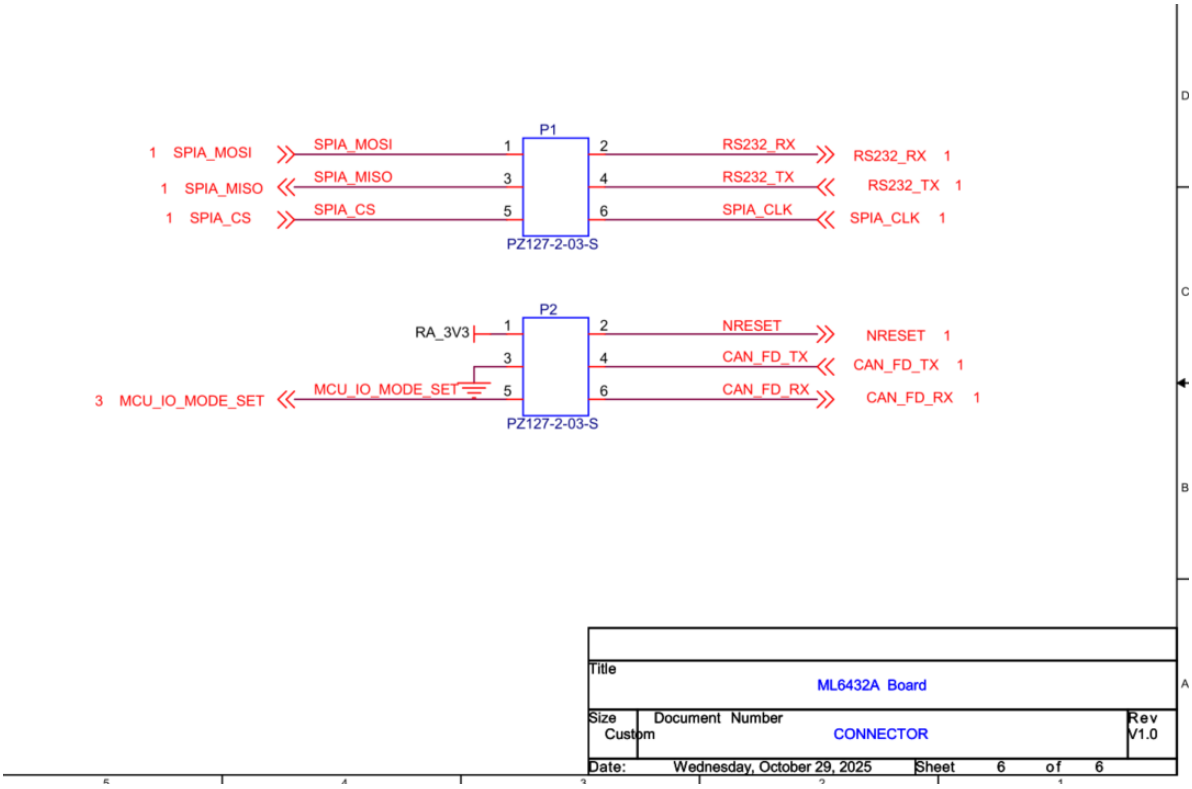


Figure 4. Interface schematic

5.1 Status LED Reference

The image below is the added ML6432A status LED reference, which can be used to quickly locate the on-board indicator position.

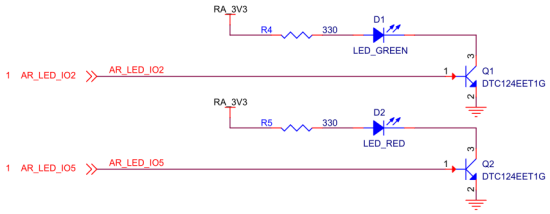


Figure 5. ML6432A status LED reference

6. Usage and Flashing Instructions

The module supports both MMWK flashing and standard flashing. For MMWK flashing, refer to the MMWK documentation. Before standard flashing, module debugging, or firmware programming, prepare the following drivers and tools according to the actual hardware configuration.

1 Download

1 Download

1 Download

1 Download

6.1 Flashing Reference

The image below is the added ML6432A flashing reference, which can be used as an auxiliary guide when connecting the flashing tool and programming firmware.

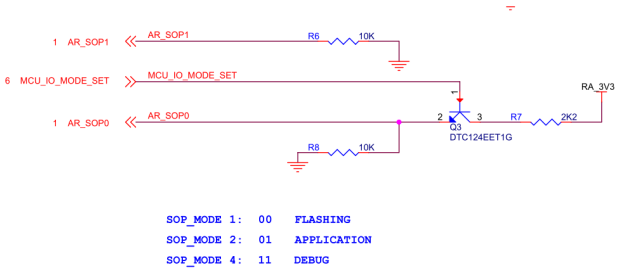


Figure 6. ML6432A flashing reference

6.2 Boot Mode Configuration

The module boot mode is controlled by the MCU_IO_MODE_SET (P2.5) pin. Different logic levels correspond to different operating states.

1 P2.5

GND

1 P2.5

3.3V

10kΩ

6.3 Firmware Flashing Procedure

1 Configure the device for flashing mode.

1 Connect the device to the computer through the serial tool.

1 UniFlash

For detailed operating steps, refer to the official TI document: Documentation

6.4 Module Usage Instructions

After the firmware is programmed:

1 Switch the device to application boot mode.

1 Connect to the device through the serial port.

1 Use the serial tool to view runtime data, send configuration files, or issue debugging commands.

6.5 Serial Port Connection

The hardware connection is as follows.

Device Port	Serial Tool
3V3	3.3V output
GND	GND
RS232_RX	TX
RS232_TX	RX

When the serial debugging adapter is connected to the computer through `USB`, the system recognizes it as a serial device. The assigned port name depends on the host system. On `Windows`, it is typically shown as a `COM` port such as `COM20`.

7. Related Documents

1 MDR Module Introduction

1 WDR-M Main Controller Carrier Board Introduction

1 ML6432A_BO Module Introduction